

Grade 3 Ontario Curriculum Data

Data Detectives — Grade 3 Case

Unit Overview: Data Detectives — Grade 3 Case

Grade: Grade 3 | **Strand:** Data

Target Expectations: D1.1, D1.2, D1.3, D1.4, D1.5

Learning Goal: Children will sort by 3 attributes, build frequency tables, use many-to-one scale on graphs, calculate mean and mode, and analyse data with conclusions.

Lesson	Lesson Title	Learning Goal
Day 1	Three-Attribute Sort	I can sort by 3 attributes using a tree diagram.
Day 2	Frequency Tables	I can build a frequency table with qualitative or quantitative data.
Day 3	Many-to-One Bar Graphs	I can use many-to-one scale on a bar graph.
Day 4	Mean and Mode	I can find the mean and mode of a data set.
Day 5	The Big Case Capstone	I can analyse data and write a conclusion using mean and mode.

Lesson 1: Three-Attribute Sort

Goal: Tree diagrams sort by 3 attributes by branching at each step.

Expectation: D1.1

Learning Goal: "I can sort by 3 attributes using a tree diagram."

Materials: M1_tree_mat, M5_data_cards, M7_data_chart, M10_unit_poster

1. Minds On (12 mins): Detective Dot Branches the Sort

Detective Dot: 3 attributes! RED or BLUE? SQUARE or CIRCLE? BIG or SMALL? Each is a BRANCH on the tree! Dot opens: 'Three levels, three rules — same rule on every BRANCH!'

Bring children to the carpet. Detective Dot and Tally puppets. 'Friends — last year you sorted by 2 attributes with Venn and Carroll. Today we sort by 3 attributes with a TREE DIAGRAM.'

2. Action (25 mins): Tree Stations

- Whole group: build a class tree with 8 leaves.
- Three stations.
- Stretch at C: explain why 3 attributes give 8 leaves.
- Discuss when tree vs Venn vs Carroll.
- Circulate to record diagnostic evidence.
- STRATEGY COMPARE: After Station C, two pairs share trees with DIFFERENT attribute orders (colour-shape-size vs size-shape-colour).

3. Consolidation (13 mins): Detective Dot's Tree Brag

Discuss:

- What's a TREE DIAGRAM?
- How many leaves in a 3-attribute tree?
- When is tree better than Venn?
- What would 4 attributes give?
- Dot's preview: tomorrow Tally turns these branches into a FREQUENCY TABLE (Lesson 2 — D1.1).

Exit: Detective Dot's branch refrain — 'Three levels, three rules — same rule on every BRANCH!' (3x).

Lesson 2: Frequency Tables

Goal: FREQUENCY TABLES list categories with their COUNTS. Qualitative = words. Quantitative = numbers.

Expectation: D1.2

Learning Goal: "I can build a frequency table with qualitative or quantitative data."

Materials: M2_frequency_template, M5_data_cards, M7_data_chart, M10_unit_poster

1. Minds On (12 mins): Tally Builds a Frequency Table

Tally: Favourite drink — qualitative WORDS. Number of pets — quantitative NUMBERS. Both go in a FREQUENCY TABLE! Tally opens: 'WORDS go in QUAL, NUMBERS go in QUANT!'

Bring children to the carpet. Detective Dot and Tally puppets. 'Friends — yesterday tree diagrams. Today FREQUENCY TABLES.'

2. Action (25 mins): Frequency Stations

- Whole group: build a frequency table for 'favourite season' (8 students).
- Three stations.
- Stretch at C: invent 3 of each kind.
- Discuss qualitative vs quantitative.
- Circulate to record formative evidence.
- INVENT-AND-COMPARE: Pairs design TWO different frequency table layouts for the same survey.

3. Consolidation (13 mins): Tally's Frequency Brag

Discuss:

- What's a FREQUENCY TABLE?
- Qualitative vs quantitative?
- Give one of each.
- Why do we tally before counting?

Exit: Tally's table refrain — 'Tally's rule: WORDS go in QUAL, NUMBERS go in QUANT!' (3x).

Lesson 3: Many-to-One Bar Graphs

Goal: On bigger data, ONE bar height = TWO or FIVE or TEN units. SCALE matters!

Expectation: D1.3

Learning Goal: "I can use many-to-one scale on a bar graph."

Materials: M3_bar_grid, M8_scale_chart, M7_data_chart, M10_unit_poster

1. Minds On (12 mins): Detective Dot Picks a Scale

Detective Dot: 30 students like hockey. The bar would be HUGE! Let's use SCALE — 1 unit = 5 students. Bar = 6 units! Dot opens: 'CHECK the scale, MULTIPLY the height — that's the COUNT!'

Bring children to the carpet. Detective Dot and Tally puppets. 'Friends — yesterday frequency tables. Today MANY-TO-ONE SCALE.'

2. Action (25 mins): Scale Stations

- Whole group: scale a 30-student data set.
- Three stations.
- Stretch at C: explain why scale matters.
- Discuss why scale matters.
- Circulate to record formative evidence.
- INVENT-AND-COMPARE: At Station C, pairs design TWO different scales for the same 50-student data (1 unit = 5 vs 1 unit = 10).

3. Consolidation (13 mins): Detective Dot's Scale Brag

Discuss:

- What's MANY-TO-ONE?
- Why pick a scale?
- How do we read a scaled bar?
- Where do we mark scale?

Exit: Detective Dot's scale refrain — 'CHECK the scale, MULTIPLY the height — that's the COUNT!' (3x).

Lesson 4: Mean and Mode

Goal: MEAN = total ÷ count (the 'fair share'). MODE = most common.

Expectation: D1.4

Learning Goal: "I can find the mean and mode of a data set."

Materials: M4_mean_chart, M9_calculator_set, M7_data_chart, M10_unit_poster

1. Minds On (12 mins): Tally Calculates Mean

Tally: 5 friends ate cookies: 2, 4, 3, 5, 6. Total = 20. Divide by 5 friends. MEAN = 4 cookies each! Tally opens: 'ADD 'em up, SHARE 'em out!'

Detectives — yesterday we used SCALE on bar graphs (1 unit = 5). Today we COMPUTE the mean and find the mode. Both summarize a data set, but in different ways.

2. Action (25 mins): Mean Stations

- INVENT-A-STRATEGY: Before showing the formula, give 4 children with 2, 4, 6, 8 counters.
- Whole group: calculate mean of 5 small data sets.
- Three stations.
- Stretch at C: explain when mean tells more than mode.
- Discuss when mean $\neq$ mode.
- Circulate to record formative evidence.

3. Consolidation (13 mins): Tally's Mean Brag

Discuss:

- What's MEAN?
- What's the formula?
- When is mean DIFFERENT from mode?
- Which is more useful?

Exit: Tally's two-step refrain — 'Tally's Two-Step: ADD 'em up, SHARE 'em out!' (3x).

Lesson 5: The Big Case Capstone

Goal: Detectives use sort, frequency, graphs, mean+mode to TELL the data's story.

Expectation: D1.5

Learning Goal: "I can analyse data and write a conclusion using mean and mode."

Materials: M1_tree_mat, M2_frequency_template, M3_bar_grid, M4_mean_chart

1. Minds On (12 mins): Detective Dot's Big Case

Detective Dot: Today's BIG CASE — survey, table, scale graph, mean, mode, conclude. The complete detective sweep! Dot + Tally open: 'Sort, TABLE, GRAPH, MEAN/MODE — case CLOSED!'

Bring children to the carpet. Detective Dot and Tally puppets in capes. 'Friends — today is our CAPSTONE. We use ALL we know: tree diagrams, frequency tables, scaled graphs, mean, mode, and conclusions.'

2. Action (25 mins): The Big Case

- Hand out templates.
- Stations 1-5: tree, frequency, scale, mean+mode, conclude.
- Stretch: write a 2-sentence conclusion.
- Help with mean calculation as needed.
- Circulate to record summative evidence.
- STRATEGY COMPARE: Two pairs share their case analysis — Pair A focused on mode, Pair B on mean.

3. Consolidation (13 mins): Detective Dot's Big Case Brag

Discuss:

- What's MEAN?
- What goes in a frequency table?
- What's many-to-one scale?
- What's a CONCLUSION with mean and mode?

Exit: Capstone refrain — Dot + Tally call: 'Sort, TABLE, GRAPH, MEAN/MODE — case CLOSED!' Class chants 3x.

Name: _____

Date: _____

Detective Dot's Tree Sort

Learning Goal: I can sort by 3 attributes using a tree diagram.

Instructions for the Student:

1. Place 8 items at the right leaves of the tree (colour x shape x size).
2. Read the pre-built tree.
3. Pick 3 attributes for your toys.

Make Your Own Frequency Table

Item	Tally	Frequency

Name: _____

Date: _____

Tally's Frequency Builder

Learning Goal: I can build a frequency table with qualitative or quantitative data.

Instructions for the Student:

1. Sort 6 survey questions as QUAL (words) or QUANT (numbers).
2. Use the tally chart to build a frequency table.
3. Make 1 qualitative + 1 quantitative survey question.

Sort and Categorise

Sort the data and write the categories:

Name: _____

Date: _____

Detective Dot's Scale Lab

Learning Goal: I can use many-to-one scale on a bar graph.

Instructions for the Student:

1. Look at the bar graph.
2. Use the frequency table to build a scaled bar graph.
3. For 3 different data sets (counts shown), pick a scale.

Pick the right SCALE

scale = 1

Small data (≤ 10)

scale = 5

Medium (10–50)

scale = 10 or 20

Large (50+)

Make the tallest bar fit on the page.

Tip

Name: _____

Date: _____

Tally's Mean Lab

Learning Goal: I can find the mean and mode of a data set.

Mission

1. Calculate the mean for 5 data sets.
2. Find the MODE for 6 data sets.
3. Calculate BOTH mean and mode for 3 data sets.

Find the MEAN (average)

Goals scored: $2 + 3 + 4 + 5 + 6$
Mean = $2 + 3 + 4 + 5 + 6 / 5 =$

Pages read: $10 + 12 + 15 + 18$
Mean = $10 + 12 + 15 + 18 / 4 =$

Hours slept: $8 + 9 + 7 + 8$
Mean = $8 + 9 + 7 + 8 / 4 =$

Name: _____

Date: _____

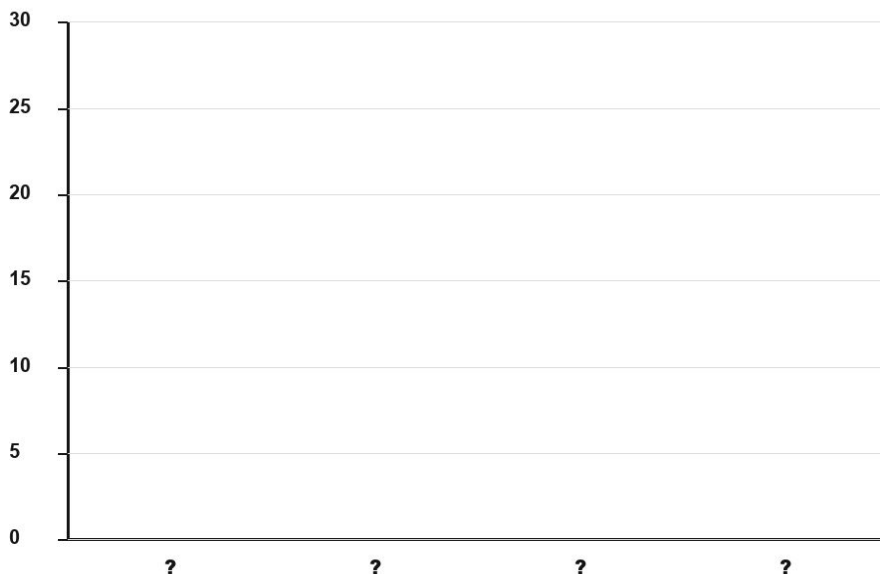
Detective Dot's Big Case

Learning Goal: I can analyse data and write a conclusion using mean and mode.

Instructions for the Student:

1. Build a frequency table from the survey data.
2. Build a bar graph (1 unit = 2).
3. Write a conclusion using BOTH mean and mode.

Capstone Bar Graph

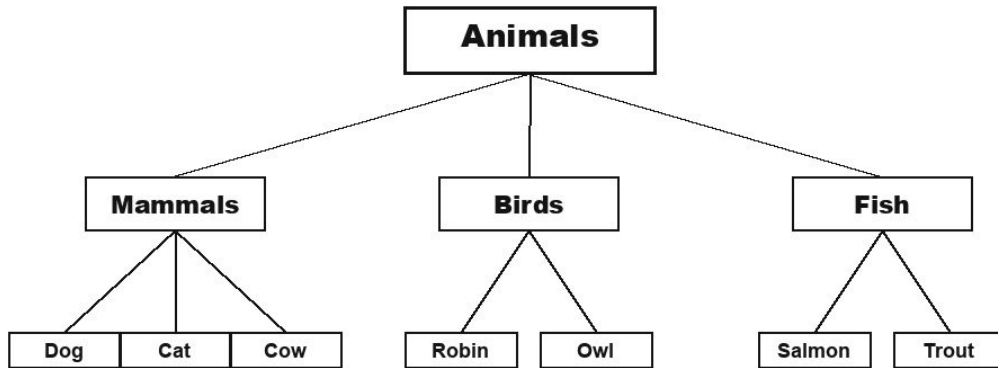


Tree Diagram Mat

(M1_tree_mat)

Letter-size mat with branching tree for 3-attribute sorting.

Tree Diagram



PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 12 mats.

2. Laminate for dry-erase.

3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 mats

Lessons: 1, 5

Prep: ~15 min

Frequency Table Template

(M2_frequency_template)

Letter-size template for frequency tables.

Frequency Table

Category	Tally	Frequency
Pizza		8
Pasta		6
Salad		4

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 12 templates.

2. Laminate.

3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 templates

Lessons: 2, 5

Prep: ~10 min

Scaled Bar Graph Grid

(M3_bar_grid)

Letter-size bar grid with adjustable scale.

My Pictograph

PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 12 grids.
2. Laminate.
3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 grids
Lessons: 3, 5
Prep: ~10 min

Mean Calculation Chart

(M4_mean_chart)

Anchor chart showing mean = total ÷ count.

Find the MEAN (average)

Set A: $4 + 6 + 8$

Mean = $4 + 6 + 8 / 3 =$

Set B: $3 + 5 + 5 + 7$

Mean = $3 + 5 + 5 + 7 / 4 =$

Set C: $10 + 12 + 14 + 16$

Mean = $10 + 12 + 14 + 16 / 4 =$

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall before Lesson 4.

USE IN CLASS

Quantity: 1 chart

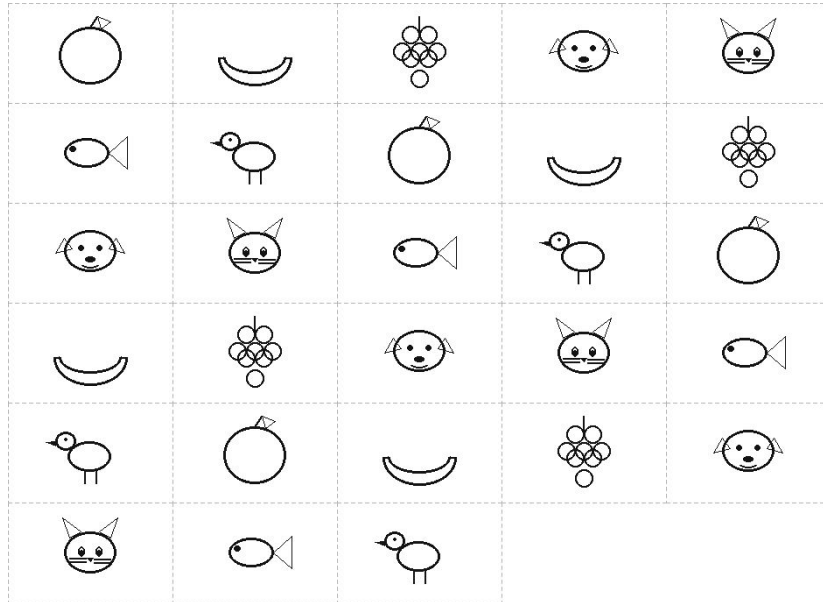
Lessons: 4, 5

Prep: ~10 min

Survey Data Cards

(M5_data_cards)

30 paper data cards with 3 attributes each.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 30 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 2, 5

Prep: ~25 min

Big Case Template

(M6_capstone_template)

Letter-size capstone template.

1. My QUESTION:
2. My DATA (tally):
3. My PICTOGRAPH:
4. What I FOUND:
5. My CONCLUSION:

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

TEACHER PREP

1. Print 25 templates.

2. Stack near teacher table.

3. No lamination.

USE IN CLASS

Quantity: 25 templates

Lessons: 5

Prep: ~5 min

G3 Data Vocabulary Chart

(M7_data_chart)

Anchor chart with G3 data vocabulary.

Data Words	
DATA	Information we collect to answer a question.
TALLY	Marks we use to count. Five marks = a bundle.
PICTOGRAPH	A graph that uses pictures to show data.
CATEGORY	A group that things belong to.
MOST	The category with the largest number.
LEAST	The category with the smallest number.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall before Lesson 1.

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

Scale Chart

(M8_scale_chart)

Anchor chart explaining many-to-one scale.

SCALE	
SCALE	How much each unit on the axis represents.
Why	Lets us show big data on a small graph.
Example	Scale of 5 → each square = 5 students.
Tip	Pick a scale that fits the largest value.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape next to vocab chart.

USE IN CLASS

Quantity: 1 chart

Lessons: 3, 5

Prep: ~10 min

Paper Calculator Cards

(M9_calculator_set)

12 paper calculator cards for mean calculation.

MEAN, MEDIAN, MODE, RANGE

MEAN

Add all, divide by how many.

MEDIAN

The middle value (in order).

MODE

The value that appears most often.

RANGE

Highest minus lowest.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 12 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 4, 5

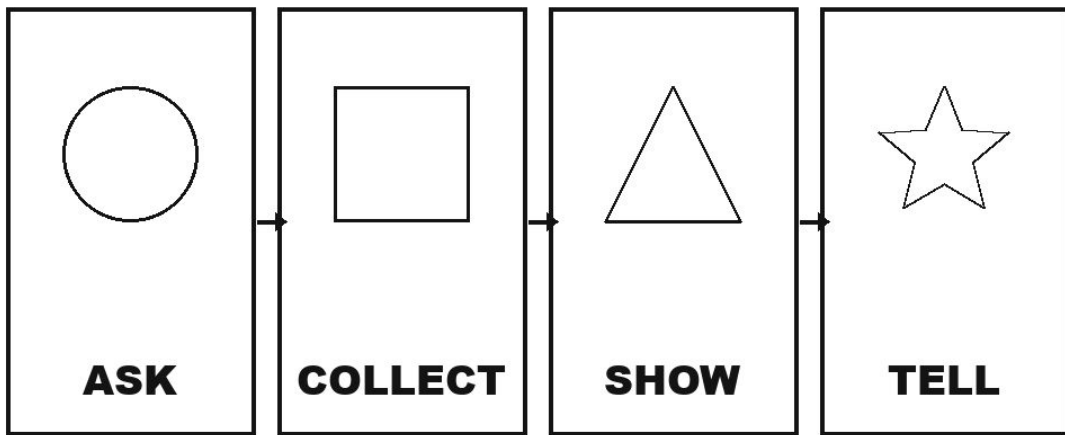
Prep: ~15 min

G3 Detectives Poster

(M10_unit_poster)

Unit anchor poster of Dot and Tally.

Data Detectives!



Ask. Collect. Show. Tell.

PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 1 copy on letter or 11x17.
2. Laminate.
3. Tape to front wall on Day 1.

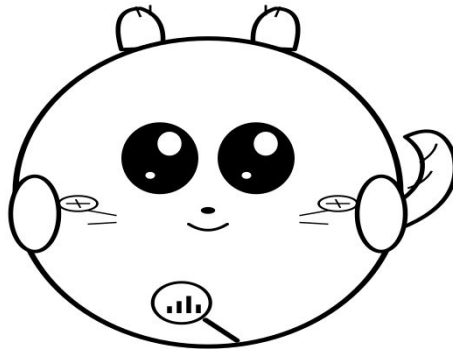
USE IN CLASS

Quantity: 1 poster
Lessons: 1, 2, 3, 4, 5
Prep: ~10 min

Detective Dot Puppet

(char_dot_puppet)

Paper puppet of Detective Dot for teacher use in all 5 lessons.



Detective Dot the Squirrel

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette + tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

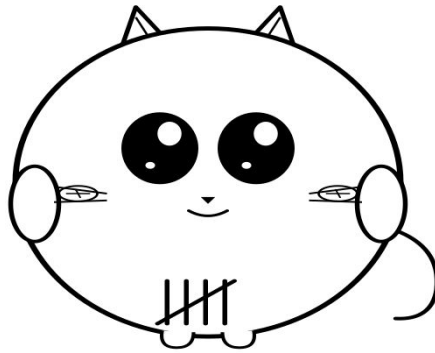
Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

Tally Puppet

(char_tally_puppet)

Paper puppet of Tally for teacher use in Lessons 2-5.



Tally the Tally Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette + tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~12 min

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
D1.1 sort sets of data about people or things according to two and three attributes, using tables and logic diagrams, including Venn	Cannot use tree diagram. Mixed branches at same level.	Builds tree with prompts. Branches mostly consistent.	Independently builds 3-attribute tree with consistent branches.	Plans 4+ attribute tree and explains 2^N leaves.
D1.2 collect data through observations, experiments, and interviews to answer questions of interest that focus on qualitative and	Mixes qualitative and quantitative.	Builds frequency table with prompts.	Independently asks question, distinguishes qual/quant, builds frequency table.	Articulates qual/quant difference and uses both data types.
D1.3 display sets of data, using many-to-one correspondence, in pictographs and bar graphs with proper sources, titles, and labels	Reads scaled bars as 1-to-1.	Reads scale with prompts.	Independently picks scale and reads/builds scaled bars.	Justifies scale choice and adapts scale for different data sets.
D1.4 determine the mean and identify the mode(s), if any, for various data sets involving whole numbers, and explain what each of	Wrong divisor for mean. Cannot find mode.	Calculates mean with prompts.	Independently calculates mean and identifies mode (including no-mode and bimodal).	Explains when mean $\neq$ mode and which is more useful.
D1.5 analyse different	Vague conclusion without measures.	Conclusion with one measure.	Independently writes conclusion using	Uses conclusions to make informed

Junior Data Detective Certificate (G3)

This certificate is proudly presented to:

For successfully completing the 5-Day G3 Data Detectives unit and demonstrating the skills of a G3 Junior Data Detective. Detective Dot the Sorting Squirrel congratulates the student.

By completing this unit, this student has demonstrated:

- Sorting by 3 attributes with tree diagrams
- Building frequency tables (qualitative + quantitative)
- Using many-to-one scale on bar graphs
- Calculating mean and finding mode

Marketplace Listing

A 5-day Grade 3 unit on data framed around Detective Dot and Tally.

Price: \$8.99 CAD

Pages: ~45

Classroom time: 250 min

Teacher prep: ~200 min

Top tags: Grade 3, math, data, tree diagram, frequency table, bar graph, mean, mode

What's included:

- Unit blueprint (1 file)
- 5 lesson plans (5 files)
- 5 worksheets (5 files)
- Manipulatives plan (1 file)
- Formative + reflection (1 file)
- Assessment suite (1 file)